

# NBA Scouting: Player Analysis and Suggestions for Recruitment

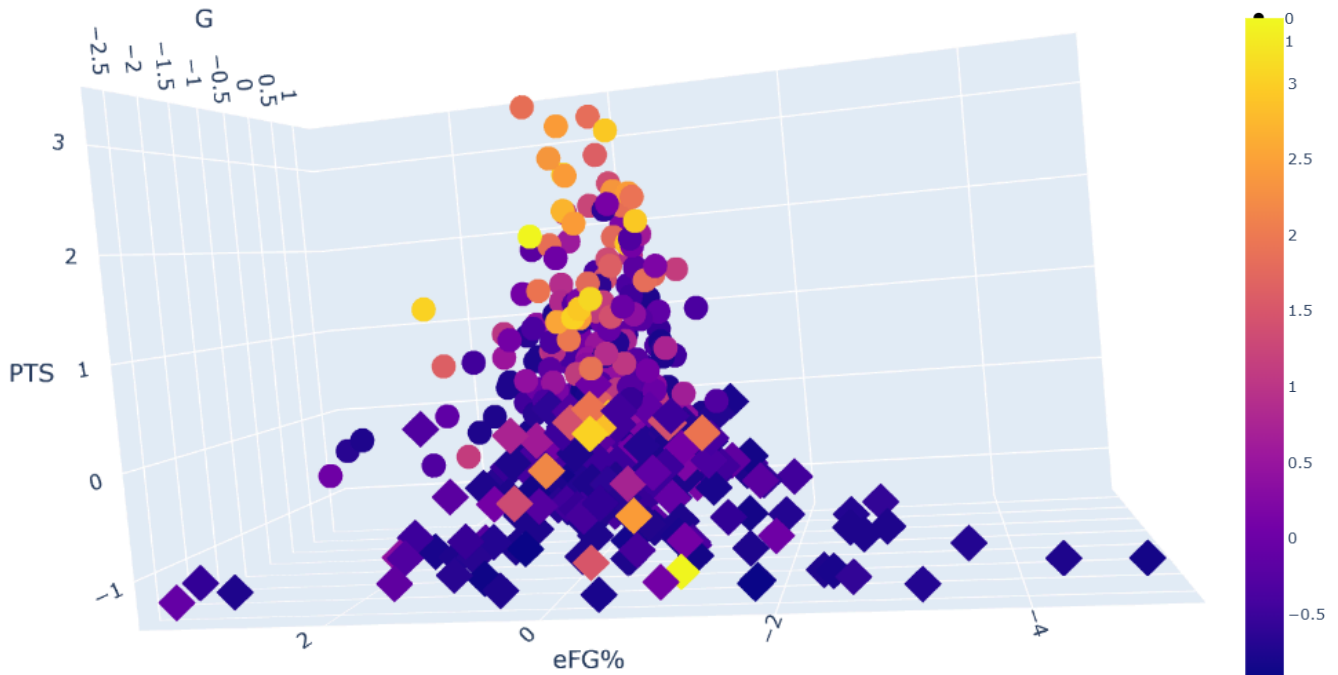


Figure 1: 3D scatterplot of players based on effective field goal percentage (eFG%), games played (G), and points scored (PTS) with colors indicating salary (light=high, dark=low)

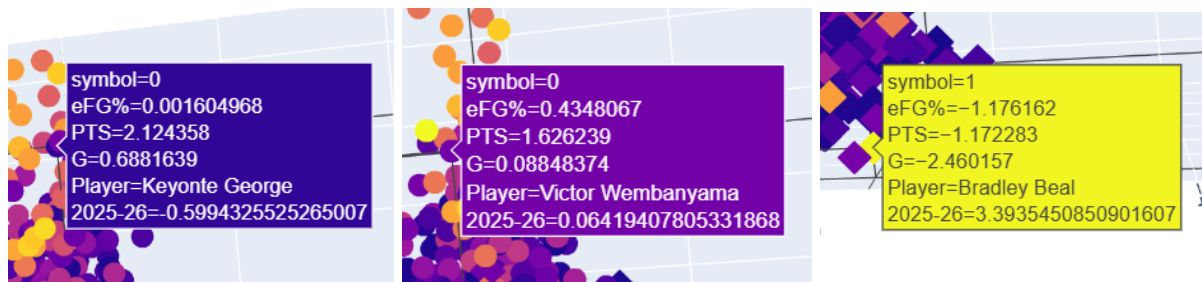


Figure 2: Example players - Good choices vs. Alternative choices vs. Bad choices

What is this visualization and what does it mean? (Figure 1)

For scouting out players that are high performing but not highly paid, our analytics team decided to run a classification algorithm called kMeans. This model used player data (a documentation of each player's rank, age, games played, 3-point field goal percentage, 2-point field goal percentage, effective field goal percentage, free throw percentage, total rebounds, points scored, triple-double) to classify players into two groups: high-performing and low-performing. We then took the outputs and graphed them in a 3D scatterplot, where each point represents one player (note: if a player played in multiple teams, we took only the data from the team that the individual played the highest number of games in). If the point is a circle, this indicates that the player is high-performing. If the point is a square, this indicates that the player is low-performing. Lighter colors indicate that the player is highly paid, while dark colors indicate that the player is not paid as highly. We determined that the three variables that contributed the most to differentiating high-performing players and low-performing players were effective field goal percentage (eFG%), games played (G), and points scored (PTS), which were ultimately plotted against the horizontal, vertical, and depth axes of the graph respectively.

How is this visualization used to determine which players to scout? (*Figure 2*)

When scouting new players, we want individuals that perform well but are underpaid. Using the 3D scatterplot, we will want to look into players who are represented as circles with a dark color, with key examples being Keyonte George, Shaedon Sharpe, and Kon Knueppel. This is due to the fact that they are classified as being high-performing players (circles) but are being paid less than the average player (dark color). Keyonte George's stats prove this, as he has an extremely high standardized PTS of 2.124, which indicates that he is at the 98th percentile of all players. His standardized eFG% is about average and his standardized G is at the 75th percentile. However, he has a low standardized salary of -0.599, which is roughly at the 27th percentile of all players.

If we cannot recruit these players, we will want to look into players that are represented as circles with a slightly lighter color, which would indicate they are high-performing but are being paid at the average amount. Key examples for alternative options are Victor Wembanyama, Devi Avdija, and Cooper Flagg. Victor Wembanyama's stats show that he has a very high standardized PTS of 1.626, which indicates that he is at the 95th percentile of all players. His standardized eFG% is at the 67th percentile and his standardized G is about average. His standardized salary is about average.

Players that we are excluding from our potential recruits are individuals that perform poorly and are overpaid. This can be determined by looking into players that are represented as squares with a very light color, with key examples being Bradley Beal, Anthony Davis, and Trae Young. This is due to the fact that they are classified as being low-performing players (squares) but are being paid more than the average player (light color). Bradley Beal's stats prove this, as he has negative standardized eFG%, G, and PTS, indicating below average stats, but his standardized salary is 3.394, which is roughly above the 99th percentile of all players, indicating extremely high pay for low performance.